NARAYANA ENGINEERING COLLEGE

An Autonomous College under JNTUA, Approved by AICTE, New Delhi

Dhurjati Nagar, Gudur, Tirupati Dist., AP - 524101

Department of Computer Science & Engineering

Freshmen Engineering Department

MODEL QUESTIONS

S.NO	Questions	Marks
1	What are the characteristics of a computer? Ans: Speed, Accuracy, Diligence, Versatility, Reliability, Consistency, Memory, Storage Capacity, Remembrance Power, and Automation.	2
2	What is a Program and Programming? Ans: A *program* is a set of logically related instructions that is arranged in a sequence that directs the computer in solving a problem. The process of writing a program is called *programming*.	2
3	Define Software and hardware? ***Ans:Software is*** a set of instructions, data or programs used to operate computers and execute specific tasks***.*** ***Hardware refers to*** the computer's tangible components or delivery systems that store and run the written ***instructions provided by the software.***	2
4	What is program counter? Ans : The program counter (PC) is a register that manages the memory address of the instruction to be executed next.	2
5	Write any two advantages of writing algorithms. Ans: 1. It is not dependent on any programming language, so it is easy to understand for anyone even without programming knowledge. 2. Every step in an algorithm has its own logical sequence so it is easy to debug. 3. By using algorithm, the problem is broken down into smaller pieces or steps hence, it is easier for programmer to convert it into an actual program.	2
	Differentiate between Compiler and Interpreter.	

6	Ans : 1. Compilers scan the entire program in one go. Interpreter: The program is interpreted/translated one line at a time. 2. Compilers convert the source code to object code. Interpreters do not convert the source code into object code.	2
7	What is Pseudocode? Ans : Pseudo-code is a written statement of an algorithm using a restricted and well-defined vocabulary. It is similar to a 3GL, and for many programmers and program designers it is the preferred way to state algorithms and program specifications.	2
8	Differentiate RAM and ROM. Ans: 1. RAM stands for Random Access Memory. ROM stands for Read Only Memory. 2. RAM is a volatile memory. Which means it stores data as long as power is on. ROM is a non-volatile memory that means it stores data permanently even when power is off.	2
9	What is a Keyword? Write any 4 C keywords. Ans: Keyword is a reserved word. We cannot use it as a variable name, constant name, etc. There are only 32 reserved words (keywords) in the C language. Ex: auto, double, int, float	2
10	What is an Identifier? Write any two rules of naming C identifiers. Ans: Identifier refers to name given to entities such as variables, functions, structures etc. ● Rules for naming identifiers 1. A valid identifier can have letters (both uppercase and lowercase letters), digits and underscores.	2

	2. The first letter of an identifier should be either a letter or an underscore.	
11.	What is variable declaration? Ans : Declaration of a variable in a computer programming language is a statement used to specify the variable name and its data type. Syntax: datatype variablename; Ex: int a;	2
12.	What is variable initialization? Ans : Initialization is the process of assigning a value to the Variable. Synatx: datatype variablename=value; Ex: int a=5;	2
13.	Differentiate formatted and unformatted I/O. Ans : Formatted and Unformatted I/O operations are Input/Output operations that are utilized in C programming. Unformatted I/O is used to read and write data as a stream of bytes without any format, whereas formatted I/O is used to read and write data in a predefined format.	2
14.	Explain getchar( ) and putchar( ) functions. Ans: int getchar() = Returns the next character from the standard input (stdin). Basically read one character at a time. putchar(int) = Writes a character to the standard output (stdout)	2
15.	Explain ternary operator in C. Ans: We use the ternary operator in C to run one code when the condition is true and another code when the condition is false. For example, (age >= 18) ? printf("Can Vote") : printf("Cannot Vote");	2
	What is Operator precedence and Associativity?	

16.	Ans: Precedence is the priority for grouping different types of operators with their operands. Associativity is the left-to-right or right-to-left order for grouping operands to operators that have the same precedence.	2
17.	What is Explicit Type Conversion? Ans: Explicit type conversion, also called type casting, is a type conversion which is explicitly defined within a program. Ex: a=int(b);	2
18.	Evaluate e = a + (b * c) / d if a = 20; b = 10; c = 15; d = 5. Ans: 20+(10*15)/5 20+150/5 20+30 50	2
19.	List the steps involved in Problem Solving. Ans: Problem definition Problem Analysis Design Coding Testing Maintenance	2
	Differentiate top down and bottom-up approaches. **Ans:** **Top-Down Approach** is an approach to design algorithms in which a bigger problem is broken down into smaller parts. Thus, it uses the decomposition	

20.	approach. This approach is generally used by structured programming languages such as C, COBOL, FORTRAN. **Bottom-Up Approach** is one in which the smaller problems are solved, and then these solved problems are integrated to find the solution to a bigger problem. Therefore, it uses composition approach. **Ex: C++, Java and Python**	2

MODULE-I				
Introduction to Programming and Problem Solving				
S.NO				**Marks**
1	What is a Computer? Explain the basic Organization of a computer with neat diagram.			10
2	What is a program? Explain types of programming languages.			10
3	a) What is an algorithm? Explain characteristics of an algorithm. b) Write an algorithm to exchange the values of two variables using third variable.			5 5
4	a) What is a flow chart? Explain various symbols used in flow chart. b) design a flow chart to find whether a given number is even or odd.			5 5
5	Explain the compilation and execution process.			10
6	Explain the basic structure of a C program with a diagram.			10
7	What is a data type? Explain various data types in c with examples.			10
8	What is a constant? Explain different types of C constants with examples.			10
9	a) What is a Variable? Explain the rules for naming the C variables**.** b) Write a C program to add given two			5 5

	integers and display the result.			
10	Explain about the basic Input and Output functions in C.			10
11	Explain different C Operators with examples.			10
12	a) Explain about Type Conversion and Type Casting. b) Explain Time and Space complexities of algorithms.			5 5
13	a) Explain generations of computer. b) Differentiate top down and bottom up problem solving strategies.			5 5

MODULE-II

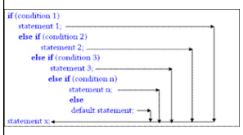
Control Structures

S.NO	Questions			Marks
1	What are Simple sequential programs? Ans: C programs are sequential in nature. Generally, sequence of statements are written in order to accomplish a specific activity. So statements are executed in the order they are specified in the program. This way of executing statements sequentially is known as Sequential control statements.			2
2	What is a Simple if Statement? Ans: The if in C is a decision-making statement that is used to execute a block of code based on the value of the given expression. If the given condition is true then the statements inside the body of “if” will be executed. If the condition is false then the statements inside the body of “if” will be skipped.			2
3	Draw Flowchart to show the working principle of if-else statement. Ans: 			2
4	What is the use of else statement in C? Ans : If you require code to run only when the statement returns true (and do nothing else if false) then an else statement is not needed. On the other hand if you need a code to execute “A” when true and “B” when false, then you can use the if / else statement.			2

5	Write a C program to find the greatest among given two integers.
	
	Ans:
	
	#include<stdio.h>
	
	int main ()
	
	{
	
	int num1, num2;
	
	num1=12,num2=13;
	
	if (num1 == num2)
	
	printf("both are equal");
	
	else if (num1 > num2)
				2

		printf("%d is greater", num1);				

	6	Write a C program to print the ASCII Value of a given Character.
		
		Ans :
		
		#include <stdio.h>
		
		int main() {
		
		char c;
		
		printf("Enter a character: ");
		
		scanf("%c", &c);
		
		// %d displays the integer value of a character
		
		// %c displays the actual character
		
		printf("ASCII value of %c = %d", c, c);
		
		return 0;
		
		}
					2	

	7	List out the multi-path decision statements in C. Ans : nested if else Else if ladder and switch			2	
	8	Write the syntax of else-if ladder. Ans:  			2	
	9	What is the use of break keyword in Switch statement? Ans: The break statement to end processing of a particular labeled statement within the switch statement. It branches to the end of the switch statement. Without break , the program continues to the next labeled statement, executing the statements until a break or the end of the statement is reached.			2	
	10	When is the default block executed in Switch statement? Ans: The default statement is executed if no case constant-expression value is equal to the value of expression . If there's no default statement, and no case match is found, none of the statements in the switch body get executed. There can be at most one default statement.			2	
	11.	What are the allowable case label types in Switch statement? Ans : The value of the expressions in a switch-case statement must be an ordinal type i.e. integer, char, short, long, etc. Float and double are not allowed.			2	

	12.	Explain infinite loops in C? Ans : An infinite loop is a looping construct that does not terminate the loop and executes the loop forever. It is also called an indefinite loop or an endless loop. It either produces a continuous output or no output			2	
	13.	What is a nested loop in C? Ans : A nested loop means a loop statement inside another loop statement. That is why nested loops are also called “loop inside loops“. There is no specific limit to how many for loops can be nested inside another for loop			2	
	14.	Differentiate between break and continue statements in C. Ans: Break statement stops the entire process of the loop. Continue statement only stops the current iteration of the loop. Break also terminates the remaining iterations. Continue doesn't terminate the next iterations; it resumes with the successive iterations.			2	

	15.	Write a C program to print alphabets from A to Z.
		
		Ans:
		
		#include <stdio.h>
		
		int main() {
		
		char c;
		
		for (c = 'A'; c <= 'Z'; ++c)
		
		printf("%c ", c);
		
		return 0;
		
		}
					2	

	16.	Differentiate entry controlled and exit controlled loops in C. Ans: The loop in which test condition is checked in the beginning of the loop are known as entry controlled loop. For Example: while loop. Whereas when statements inside the loop body is executed and then the condition is checked that loop is known to be as exit controlled loop. Ex: do while loop.			2	

MODULE-II

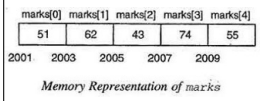
Control Structures

S.NO				**Marks**	
1	Explain the following: (i) Simple if Statement (ii) if – else Statement			5 5	
2	Explain the following: (i) Nested if Statement (ii) else if Ladder Statement			5 5	
3	Explain switch statement in C language with an example.			10	
4	Explain in detail about while Loop Statement with an example program.			10	
5	Explain in detail about do while Loop Statement with an example program.			10	

6	Explain in detail about for Loop Statement with an example program.			10	
7	Explain in detail about break and continue Statements with an example programs.			10	
8	Explain in detail about goto Statements with an example programs.			10	
9	a) Explain the working principle of while loop. b) Write a C program to find whether the given number is palindrome or not.			5 5	
10	a) Design Algorithm and Flow chart to the problem of finding whether the given number is Even or Odd. b) Write a C Program to the above said problem and show the expected Result.			5 5	

MODULE-III

Arrays and Strings

S.NO	Questions			Marks
1	What is an Array? ***Ans:*** An array is defined as the collection of similar type of data items stored at contiguous memory locations. Arrays are the derived data type in C programming language which can store the primitive type of data suchas int, char, double, float, etc.			2
2	Define and draw the memory model of a one-dimensional Array. Ans: If an array is declared, the way compiler allocates memory for the array is called memory model of array.For example if an array is declared as int marks[5];  *Memory Representation of* marks then the memory model or memory representation of such array is shown below.			2
3	What are the advantages of using arrays? Ans: The array contains the following advantages. ○ Code Optimization: Less code to access the data. ○ Ease of traversing: By using the for loop, one can retrieve the elements of an array easily. ○ Ease of sorting: To sort the elements of the array, a few lines of code only needed. ○ Random Access: Can access any element randomly using the array.			2

4	How to initialize a one-dimensional array?
	
	Ans : Initialization of array means providing the value for the array in the code when array is declared. In this method of array declaration, the compiler will allocate an array of size equal to the number of the array elements or values given.
	
	data_type array_name [size] = {value1, value2, ... valueN};
	
	Here the Size is optional, because the complier will allocate an array of size equal to the number of the array elements or values given.
	
	/* Here are a few examples */
	
	int marks[4] = { 67, 87, 56 }; // integer array initialization with four elements with default initialization of marks[3]=0
	
	float area[] = { 23.4, 6.8, 5.5 }; // float array initialization with 3 elements
				2

	What is Array traversal? Ans: Traversal is the process in which every element of the array is accessed. Generally, a for loop is used toiterate through each element of the array. Example Array Traversal using for Loop			

	5	for (int i = 0; i < N; i++)
		{
		array_name[i];
		}
					2	

	6	How to declare a two-dimensional array? Ans : The syntax to declare the 2D array is given below. For example, the 2D array with five rows, each row has three columns is : int my_array[5][3]; It declares a one-dimensional array that has five elements. Each of these five elements is themselves, an array of threeintegers.			2	
	7	Differentiate one-dimensional and two-dimensional arrays. Ans : A one-dimensional array is a single list of elements, all of which share the same data type. A two-dimensional array is an array of arrays, a list of lists , or an array of one-dimensional arrays . It displays multiple data items in list form.			2	
	8	Can we declare an array without specifying the size of an array? Ans: Yes, we can declare an array without specifying size. Syntax: - datatype arrayname[ ]={list of elements separated with comma} Ex:- int apple[ ]={5,6,7,8,9,10}; in this case compiler creates memory blocks in main memory based on number of elements we were initialized			2	

		Write a program in C to find the sum of all elements of a 1D integer array.
		Ans: #include <stdio.h>
		
		void main()
		{
		int a[100];
		int i, n, sum=0;
		
		printf("\n\Finding sum of all elements of array:\n");
		printf("----------\n");
		
		printf("Input the number of elements to be stored in the array :");
		scanf("%d",&n);				

	9	printf("Input %d elements in the array :\n",n);
		for(i=0;i<n;i++)
		{
		printf("element - %d : ",i);
		scanf("%d",&a[i]);
		}
					2	

	10	How to access the elements of an array?
		Ans: Accessing Array Elements - Array indexing is used to access the array elements using the index of the element. Since an array is stored contiguously in the memory, it has indices starting from ”0” to “array_size
		– 1. These array indices are used to access any element of the array in the following way:
		array_name[index]
		The index of the element to be accessed is specified within square brackets “[]”.
		Examples:
		int my_array[6];
		my_array[0] = 100; // access 1st element
		my_array[2] = 300; // access 4th element
		my_array[5] = 600; // access last element
					2	

	11.	How to linearly search for an element in an array? Ans : Assume KEY is the element to search. Beginning with the first element of the array, compare elements oneby one in the array with KEY element. If a match is found, then return the position in array where KEY is found. If last element is also compared but no match is found, then indicate that the KEY element is not existin the array.			2	
	12.	What is a String? Ans : The string can be defined as the one-dimensional array of characters terminated by a null ('\0'). The character array or the string is used to manipulate text such as words or sentences. Each character in the array occupies one byte of memory, and the last character must always be NULL Character. The termination character ('\0') is important as it identifies where the string ends.			2	
	13.	Why NULL character is used in String? Ans : Null character (\0) is used to terminate the string. It indicates the string ending and no processing is doneafter the identification of null character. The null character is not counted towards the length of the string but takes memory of one element. Nullcharacter cannot be read or write but is used internally only.			2	
		How to declare a String. Ans: To declare a string in C, the programmer needs to declare a one-dimensional array by specifying the type ofthe elements of array as character and the number of elements required in array as follows −				

	14.	char string_Name[size]; In the above syntax string_name is any name given to the string variable and size is used to define the lengthof the string, i.e the number of characters strings will store. Example: char city[10];			2	
	15.	Explain gets( ) function. Ans: The C standard library function, gets() can be used to take a complete line as input and store it in the string provided as a parameter. The working of gets() is such that it keeps reading the input stream until it encounters a newline character(\n). General form - gets(Array_Name)			2	
	16.	Explain any two, character testing library functions. Ans: int isalpha(int c) This function checks whether the passed character is alphabetic. int isdigit(int c) This function checks whether the passed character is decimal digit.			2	
	17.	Explain any two, character manipulation library functions. Ans:			2	

	18.	Write a C program to find the length of a string.
		
		Ans: // C program to find the length of
		
		// string using strlen function
		
		#include <stdio.h>
		
		#include <string.h>
		
		int main()
		
		{
		
		char Str[1000];
		
		printf("Enter the String: ");
		
		scanf("%s", Str);
		
		printf("Length of Str is %ld", strlen(Str));
		
		return 0;
		
		}
					2	

	19.	Explain any two, string manipulation library functions. Ans:			2	
		What is an array of strings? ***Ans:*** In C programming String is a 1-D array of characters and is defined as an array of characters. But an array of strings in C is a two-				

	20.	dimensional array of character types. Each String is terminated with a null character (\0). It is an application of a 2d array. Syntax to create a String Array: char Arrayname[rows][columns] = {"String1", "String2"...};			2	

MODULE-III

Arrays and Strings

S.NO		CO	BL	Marks	
1	What is an Array? What are the Properties of an Array? Explain the advantages of using Arrays.			10	
2	Explain Declaring and Initializing of Arrays.			10	
3	Explain Array Indexing			10	
4	Write a C program to Search an element the Single dimensional Array Elements.			10	
5	Explain Two-Dimensional arrays with suitable examples.			10	
	Explain (i) One-Dimensional arrays (ii) Two-Dimensional			05	

6	arrays.			05	
7	What is a String? Explain Declaring and Initializing of Strings with examples.			10	
8	Explain memory model of 1D and 2D array with example.			10	
9	Explain Printing of Strings with examples.			10	
10	Explain String manipulation functions			10	

MODULE-IV				
Pointers & User Defined data types				
S.NO	**Questions**	**CO**	**BL**	**Marks**
1	What are the advantages of pointers in C? Ans: Pointers in C programming are helpful to access a memory location. Pointers are an effective way to access the array structure elements. Pointers are used for the allocation of dynamic memory and the distribution.			2
2	What is syntax and example for declaring a pointer variable? Ans: The syntax of declaring a pointer is to place a * in front of the name. A pointer is associated with a type. Syntax: datatype *pointervariable name; Ex: int *a; float *b;			2
3	What is syntax and example for initializing a pointer variable? Ans: We generally use the (&) addressof operator to get the memory address of a variable and then store it in the pointer variable. int var = 10; int * ptr; ptr = &var; We can also declare and initialize the pointer in a single step. Synatx: datatype *pointervariable=&variable;			2
	What is the difference between referencing and dereferencing?			

4	Ans : a variable (such as number) directly references a value, whereas a pointer indirectly references a value through the memory address it stores. Referencing a value indirectly via a pointer is called indirection or dereferencing.			2
5	How to access the value of the data variable directly and indirectly? Ans: Direct Access: we can directly use the variable name to access the variable. Indirect Access: we use a pointer to access that variable. Ex: int a=5,*p; p=&a; printf("value of a using direct way is %d",a); printf("value of a using indirect way is %d",*p);			2
6	What is the purpose of a void pointer? Ans: A pointer to void means a generic pointer that can point to any data type. We can assign the address of any data type to the void pointer, and a void pointer can be assigned to any type of the pointer without performing any explicit typecasting.			2
7	What is the purpose of pointer to pointer? Ans: The pointer to a pointer in C is used when we want to store the address of another pointer. The first pointer is used to store the address of the variable. And the second pointer is used to store the address of the first pointer. That is why they are also known as double-pointers.			2
	How can we declare an array of pointers to integers? Ans: In C, a pointer array is a homogeneous collection of indexed pointer variables that are references to a memory location. It is generally used in C Programming when we want to point at multiple memory locations of a similar data type			

8	in our C program. We can access the data by dereferencing the pointer pointing to it. Syntax: pointer_type *array_name [array_size]; Here, ● pointer_type: Type of data the pointer is pointing to. ● array_name: Name of the array of pointers. ● array_size: Size of the array of pointers.			2

9	\What is the syntax for declaring a structure?
	
	Ans:
	
	The general syntax for a struct declaration in C is:
	
	struct tag_name
	
	{
	
	type member1;
	
	type member2;
	
	}
	
	Ex:
	
	struct MyStructure
	
	{
	int myNum;
	char myLetter;
	};
				2

10.	What is the syntax for creating a variable for structure? Ans: To access the structure, you must create a variable of it. Use the struct keyword inside the main() method, followed by the name of the structure and then the name of the structure variable: Create a struct variable with the name "s1": struct myStructure { int myNum;			2

	char myLetter;}; int main() { struct myStructure s1; return 0;}			

11.	What is the syntax for declaring a union?
	
	Ans:
	
	union union_name
	
	{
	
	datatype field_name;
	
	datatype field_name;
	
	// more variables
	
	};
	
	The fields of a union cannot be initialized all at once.
				2

12.	What is the syntax for initializing a structure? Ans: Initialization using Initializer List. struct structure_name str = { value1, value2, value3 }; In this type of initialization, the values are assigned in sequential order as they are declared in the structure template.			2
13.	How can we initialize a union? Ans: The union can hold only one value at a time. A union can be initialized on its declaration. Because only one member can be used at a time, only one can be initialized. To avoid confusion, only the first member of the union can be initialized.			2
	What is the syntax for creating a nested structure? Ans:			

14.	struct Date
	{
	int dd;
	int mm;
	int yyyy;
	};
	struct Employee
	{
	int id;
	char name[20];
	struct Date doj;
	}emp1;
	(OR)
	struct Employee
	{
	int id;
	char name[20];
	struct Date
	{
	int dd;
	int mm;
	int yyyy;
	}doj;
	}emp1;
				2

15.	What are the differences between structure and union? Ans:			2

16.	What are the similarities between structure and union? Ans: Both are user-defined data types used to store data of different types as a single unit. Their members can be objects of any type, including other structures and unions or arrays. A member can also consist of a bit field. Both structures and unions support only assignment = and sizeof operators. The two structures or unions in the assignment must have the same members and member types. A structure or a union can be passed by value to functions and returned by value by functions. The argument must have the same type as the function parameter. A structure or union is passed by value just like a scalar variable as a corresponding parameter. '.' operator or selection operator, which has one of the highest precedences, is used for accessing member variables inside both the user-defined datatypes.			2

MODULE-IV				
Pointers & User defined data types				
S.NO		**CO**	**BL**	**Marks**
1	a) What is a pointer? Explain declaration and initialization of a pointer in c with syntax and examples. b) Explain array of pointers.			5 5
2	a) Explain about referencing dereferencing of pointer variables with an examples. b) Explain in detail about multiple indirection in C.			5 5
3	Explain in detail about pointer arithmetic in C.			10
4	Explain in detail about array manipulation using pointers.			10
5	a. a. Why do we need structure in C language? b. What is structure? How to declare and initialize structure with syntax and example program.			5 5
6	a.Define Union? How to declare and initialization the union with an example program in 'C'			10
7	Explain structure member access operators with syntaxes and example programs.			10
8	Explain about Nesting of Structures with syntax and example program.			10

9	What are the differences between structure and union.			10
10	Explain about the following: a) Array of structures b) Self referential structures			5 5

MODULE-V				
Functions & File Handling				
S.NO	**Questions**	**CO**	**BL**	**Marks**
1	What is a Function? Ans: A function in C is a set of statements that when called perform some specific tasks. It is the basic building block of a C program that provides modularity and code reusability. The programming statements of a function are enclosed within { } braces, having certain meanings and performing certain operations. They are also called subroutines or procedures in other languages.			2

2	What are the three important functions aspects?
	
	Ans: There are three aspects of a C function. They are -
	
	• Function Declaration int addTwoIntegers(int a, int b);
	• Function Call addTwoIntegers(a,b);
	
	Function Definition
	
	int addTwoIntegers(int a, int b)
	
	{
	
	int sum=a+b;
	
	return sum;
	
	}
				2

3	What is Function declaration? Ans: A function declaration usually contains the function name , return type , and the parameter types. The following is the syntax for defining a function in C: return_type function_name(parameter_list); Here, return_type is the data type of the value that the function returns.			2

4	What is Function call? Ans : To use a defined function, that function must be called to perform the defined task. This is known as Function calling. When a program calls a function, the program control is transferred to the called function. A function can be called by specifying the function and passing the required arguments. If the function is returning avalue then it is to be received in calling function. Syntax : function_name (argument_list) Example :max(a,b) max(24,b);			2
5	What is a function argument? Ans: A function argument is like a placeholder. When a function is invoked, value is passed to the argument. This value is referred to as actual parameter or argument. The argument list refers to the type, order, and number of the arguments of a function. Arguments are optional; that is, a function may contain no arguments.			2
6	What is the use of return statement in functions? Ans: A return statement ends the execution of a function and returns control to the calling function. Execution resumes inthe calling function at the point immediately following the call. A return statement can return a value to the calling function. C language allows returning a single value from the function the type of value being returned must match with the return type of the function.			2
7	Write a function in C that accepts two integers and return the Sum of them. Ans: The pointer to a pointer in C is used when we want to store the address of another pointer. The first pointer is used to store the address of the variable. And the second pointer is used to store the address of the first pointer. That is why they are also known as double-pointers.			2

8	What are the types of functions based on arguments and return values? Ans: Functions can be differentiated into 4 types according to the arguments passed and value returns these are: ○ Function with no arguments and return value. ○ Function with no arguments and with return value ○ Function with argument and no return value ○ Function with arguments and return value.			2
9	What are library functions? Ans: C library functions are inbuilt functions that are in some common location known as the library. These functions areused to execute a particular operation and are generally preferred to obtain the predefined output. All C standard library functions are declared by using many header files. These library functions are created at the time of designing the compilers. Ex – printf(), scanf(), gets(), puts(), strcmp(), strstr() etc.			2
10.	What are the scope and lifetime of a C variable? Ans: The scope of a variable refers to which different parts of a program have access to the variable, that is, where thevariable is visible. Here the term variable refers to all C data types: simple variables, arrays, structure, pointers, etc. The lifetime of a variable refers that how long the variable persists in memory, or when the variable's storage isallocated and de-allocated.			2
11.	What are local and global variables? Ans: The variables are broadly classified as local and global based on scope. Local Variables - The variable that is declared in a function or code is called a local variable. The scope of Local variables is within the defined function only. You cannot			2

	use a local variable outside the function (in which it is declared). Global Variables - The scope of the global variable is the entire program. They are not defined inside any function and can be used in any function.			
12.	Explain automatic storage class in C. Ans: Auto storage class is used to declare automatic Variables. All variables declared in C programming are automatic by default. auto keyword is used to declare automatic variables. An automatic variable is a local variable whose lifetimeis within the code only. Its default value is garbage. Syntax: auto data_type variable_name = initial_value; Or data_type variable_name = initial value;			2
13.	Explain static storage class in C. Ans: Static storage class is used to declare Static Variables. The static variable is defined using the static keyword. Itsscope depends on the area of its declaration. If a static variable is defined within a function, it is a local variable. If declared outside the function, its scope is global. A static variable statically allocated memory to the variable and its lifetime throughout the program. It holds its value whenever a program is called. The default value of the static variable is 0. Syntax: static data_type variable_name = initial_value			2
14.	What is a File in C? How many types are C Files? Ans: A File is a collection of data stored in the secondary memory. A file can be classified into two types based on the way the file stores the data. They are as follows: 1. Text File 2. Binary File			2

15.	What are the advantages of using Files? Ans: Reusability: The data stored in the file can be accessed, updated, and deleted anywhere and anytime providinghigh reusability. Portability: Without losing any data, files can be transferred to another in the computer system. The risk offlawed coding is minimized with this feature. Storage Capacity: Files allow you to store a large amount of data without having to worry about storing everything simultaneously in a program.			2
16.	Differentiate Text and Binary Files in C. Ans: Text files are organized around lines, each of which ends with a newline character ('\n'). The source code files are themselves text files. A binary file is the one in which data is stored in the file in the same way as it is stored in the main memory for processing.			2
17.	What is File pointer in C? Ans: A file pointer is a reference to a particular position in the opened file. It is used in file handling to perform all file operations such as read, write, close, etc. FILE si the macro used to declare the file pointer variable. The FILE macrois defined inside <stdio.h> header file. Syntax of File Pointer : FILE* pointer_name;			2
18.	What is File I/O? Ans: The processing of reading from or writing to a File is known as File I/O. Reading From a File - The file read operation in C can be performed using functions fscanf() or fgets().fscanf() is used to read formatted string and variable arguments list to take input from a file. fgets() is used to Input the whole line from the file. Writing to a File - The file write operations can be performed by the functions fprintf() and fputs().fprinf() is used to write formatted string and variable arguments list to the file.			2

	fputs() is used to write a whole line in the file and a newline at the end.			
19.	Explain fopen() function in C. Ans: For opening a file in C, the fopen() function is used with the filename or file path along with the required accessmodes. If the file is opened successfully, fopen() returns a file pointer to it. Otherwise, returns NULL. Syntax : fopen (File_Name, Access_Mode) File_Name: name of the file to open. Access_Mode: Specifies for what operation the file is being opened.			2
20.	Explain ftell() and rewind() functions in C. Ans: ftell(): This function returns the value of the current pointer position in the file. The value is count from the beginningof the file. **Syntax:** ftell(fptr); Where fptr is a file pointer. rewind(): This function is used to move the file pointer to the beginning of the given file. **Syntax:** rewind(fptr); Where fptr is a file pointer.			2

MODULE-V				
	Functions & File Handling			
S.NO		**CO**	**BL**	**Marks**
1	a) What is a Function? What are advantages of using functions in C? b) Explain Function declaration with examples.			5 5
2	Explain Function definition with examples.			10
3	Explain (i) Function Declaration (ii) Function Call			5 5
4	Develop a C program by writing a function that finds the maximum of two numbers.			10
5	Explain function return types and arguments.			10
6	Explain parameter passing techniques.			10
7	Explain passing arrays to functions.			10
8	Explain storage classes in C..			10
9	a) What is a File? Explain advantages of using files. b) Explain the types of files in C.			5 5
10	Explain the Operations on Files in C.			10

11.	Explain random access to files in C.			10

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